RAG Assignment: Baseline Retrieval vs. Re-Ranking

Shubham Kumar Sunny
Code: https://github.com/ShubhamKumarSunny/rag-baseline-vs-reranking

19 October 2026

1 Introduction

In this assignment I compared two retrieval pipelines for RAG over the same PDF. The first one only uses a dense bi-encoder and gives the top 3 retrieved chunks to the LLM. The second one takes the same top 10 chunks, scores them again with a cross-encoder and gives the new top 3 to the LLM. The goal was to find queries where the first pipeline fails, check if re-ranking fixes them, and measure how much extra time it takes.

As the document I used the Digital Personal Data Protection Act, 2023 of India (21 pages, around 10,000 words, 44 sections and a Schedule with the penalties). Legal text is a good choice for this because it has many exceptions and "shall not" clauses, and the penalty amounts are in a table at the end of the Act instead of next to the rules they belong to.

2 Implementation

The text was extracted with PyMuPDF. In this PDF the section titles are printed in the margin, so I attached each margin note to its section, and I removed the page headers and the gazette header on the first page. For chunking I grouped paragraphs of the same section until about 130 words, with an overlap of one paragraph. A chunk never crosses a section boundary and each chunk starts with the section number and title, for example "[Section 21 - Disqualifications ...]". This gave 120 chunks with a median length of 112 words. The longest chunk is 213 tokens, so it fits in the 256 token limit of the embedding model.

Table 1: Models and settings used in both pipelines

Bi-encoder	sentence-transformers/all-MiniLM-L6-v2 (384 dimensions)
Vector store	FAISS IndexFlatIP on normalised vectors (= cosine similarity)
Cross-encoder	cross-encoder/ms-marco-MiniLM-L-6-v2
LLM	Gemini flash models through the API, temperature 0
Retrieval	top 10 from the bi-encoder, top 3 are given to the LLM

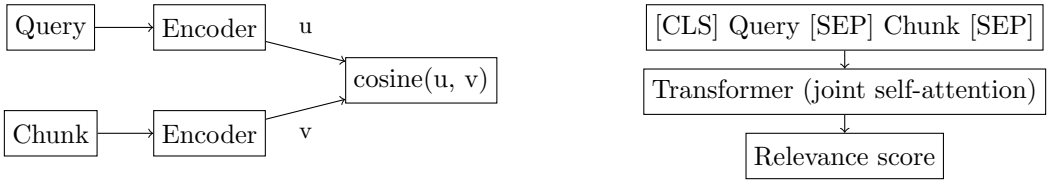
Both pipelines use exactly the same prompt. The LLM is told to answer only from the given passages, to mention the section, and to reply "The provided context does not contain the answer." if the passages do not contain the answer. All the code is in one Jupyter notebook in the GitHub repository. The LLM answers are cached in a JSON file.

3 Bi-Encoder vs. Cross-Encoder

A bi-encoder encodes the query and the chunk separately. Each of them becomes one vector (384 numbers here) and the relevance score is the cosine similarity between the two vectors. Since a chunk vector does not depend on the query, all chunk vectors can be computed once and stored in an index. For the 120 chunks this took 1.9 seconds. When a query comes in, only the query has to be encoded and then the index is searched. The disadvantage is that the model never looks at the query and the

chunk together. The whole meaning of a chunk has to fit into one vector, and small words like "not" or "except" only change that vector a little. Because of this, two chunks about the same topic get almost the same score even if one of them says the opposite of the other.

A cross-encoder gets the query and the chunk together as one input (`[CLS] query [SEP] chunk [SEP]`) and outputs one relevance score. With self-attention the query tokens and the chunk tokens attend to each other in every layer, so the model can compare them word by word. This makes it much better with negation, or with matching "not eligible" to "disqualified". The problem is that nothing can be computed in advance. For every query, each candidate chunk needs its own forward pass through the model.



(a) Bi-encoder (b) Cross-encoder

Figure 1: In the bi-encoder the query and chunk are encoded separately and only compared at the end. In the cross-encoder they go through the model together.

The difference can also be seen in the scores. In the bi-encoder top 10 the cosine scores are very close to each other. For query F8 the wrong chunk at rank 1 has a score of 0.494 and the correct chunk has 0.443. The cross-encoder scores are much more spread out and often the correct chunk is the only one with a positive score (for F8 it is 1.13 for the correct chunk and −4.18 for the next one).

4 Failure Mode Analysis

I made 10 test queries. F1–F8 are cases where the baseline fails, and L1 and L2 are cases where re-ranking does not help. For every query I first found the clause in the Act that answers it and marked the chunk that contains it as the gold chunk. I also ran a simple control query ("Can Board members be sued for actions done in good faith?"), where both pipelines had the right chunk at rank 1 and gave the same answer.

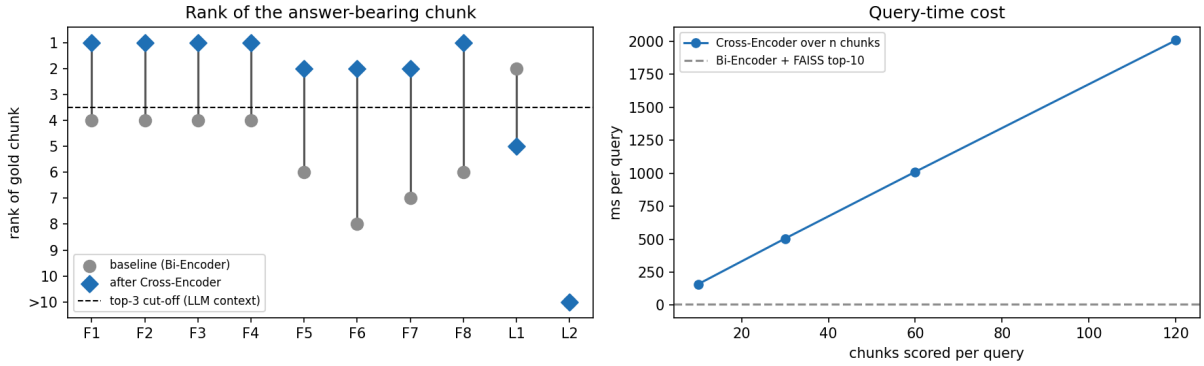


Figure 2: Left: rank of the gold chunk in the baseline (grey) and after the cross-encoder (blue). Only chunks above the dashed line are given to the LLM. Right: time per query when the cross-encoder scores n chunks directly (used in Section 6).

4.1 Keyword trap / distractor (F2, F6)

In F2 the bi-encoder put the three chunks of Section 9 on top because they are about children. But Section 9 only contains the rules. The penalty is in the Schedule, which says "Breach in observance

Table 2: Rank of the gold chunk before and after re-ranking

ID	Type	Query	Gold chunk	Before	After
F1	Negation	Who is not eligible to be a Member of the Board?	70 (s.21)	4	1
F2	Keyword trap	What is the maximum penalty for not protecting children's data?	118 (Schedule)	4	1
F3	Exception	Does the Act cover data processed for personal or domestic purposes?	10 (s.3)	4	1
F4	Buried chunk	Can the Board's order be challenged, and within how many days?	88 (s.29)	4	1
F5	Exception	Can the Central Government exempt a company from the Act entirely?	66 (s.17(5))	6	2
F6	Keyword trap	What is the punishment for a user who gives false information or files a frivolous complaint?	87 (s.28(12))	8	2
F7	Exception	Does the Act apply when a court processes personal data?	58 (s.17(1)(b))	7	2
F8	Negation	Can the salary of Board Members be reduced after they are appointed?	69 (s.20(1))	6	1
L1	Re-ranking hurts	Is parental consent always required before processing a child's personal data?	43, 44 (s.9(4))	2	5
L2	Not retrieved	Can I complain to the Board directly without first going to the company?	53 (s.13(3))	>10	>10

of additional obligations in relation to children under section 9 ... May extend to two hundred crore rupees". This chunk was at rank 4 and the cross-encoder moved it to rank 1.

F6 asks about the punishment for a user who gives false information or files a frivolous complaint. These words appear in the duties of the Data Principal (Section 15), which was ranked 2nd, and the word "punishment" matched Section 33 (Penalties) at rank 1. The actual answer is in Section 28(12): if the Board finds that a complaint is false or frivolous it "may issue a warning or impose costs on the complainant". This chunk went from rank 8 to rank 2.

4.2 Exceptions and corner cases (F3, F5, F7)

In these queries the general rule and the exception are in different chunks. In F3, Section 3 is split into two chunks. The baseline picked the second chunk of Section 3 and two chunks of Section 17, but the exception "(c) not apply to (i) personal data processed by an individual for any personal or domestic purpose" is in the first chunk. It was at rank 4 and became rank 1.

In F5 the baseline found Section 17(3), which only covers startups, and Section 35 at rank 1, which is not related to the question. The general power is in Section 17(5), where the government can declare that any provision of the Act does not apply to a Data Fiduciary. This chunk moved from rank 6 to 2. In F7 the top results were the definitions and the general "Application of Act" section. The exemption for courts and tribunals is in Section 17(1)(b), which was at rank 7 and moved to rank 2.

4.3 Negation and constraints (F1, F8)

F1 asks who is not eligible to be a Board member. The top chunk of the baseline was Section 19, which lists the qualifications for members. So it was the same topic but not what was asked. The disqualifications are in Section 21, which was at rank 4 and went to rank 1. F8 asks whether the salary of members can be reduced after they are appointed. The Act says the terms "shall not be varied to their disadvantage after their appointment" (Section 20(1)). There is almost no word overlap with the question and this chunk was at rank 6. After re-ranking it was at rank 1.

4.4 Buried chunk (F4)

F4 asks if an order of the Board can be challenged and within how many days. Chunks about the procedure of the Board (Sections 28 and 23) were ranked higher because they contain "Board" and "order" many times. The appeal section (Section 29, "within a period of sixty days") was at rank 4 and moved to rank 1. Another chunk of Section 29 also moved from rank 10 to rank 2.

4.5 Answer 1 vs. Answer 2

For all eight F queries, Answer 1 was "The provided context does not contain the answer." and Answer 2 gave the correct rule with the section number. Table 3 shows some of them.

Table 3: Examples of Answer 1 and Answer 2 (Answer 2 shortened)

ID	Answer 1	Answer 2
F2	refused	The penalty for breach of the obligations for children under Section 9 "may extend to two hundred crore rupees" (Schedule, item 3).
F5	refused	Yes, under Section 17(5) the government may, before five years from the start of the Act, declare that any provision does not apply to a Data Fiduciary.
F7	refused	Chapter II (except Section 8(1) and 8(5)), Chapter III and Section 16 do not apply when a court processes data for its judicial function (Section 17(1)(b)).
F8	refused	No, the terms of service "shall not be varied to their disadvantage after their appointment" (Section 20(1)).

So the baseline did not hallucinate in these tests. Because of the strict prompt it refused instead, which means the important clause was missing from the answer. I think with a less strict prompt the model would have tried to answer from the wrong chunks, for example by listing the qualifications from Section 19 in F1, which would be worse. In F5 and F6 the baseline context had related chunks (the startup exemption and the user duties), but they did not answer the question, so the model still refused.

Answer 2 for F6 is also not complete. It mentions the warning and costs from Section 28(12), but item 5 of the Schedule also has a penalty of up to ten thousand rupees for breaking the Section 15 duties. That chunk was at rank 53 for this query, so the re-ranker never saw it.

4.6 Where re-ranking did not help (L1, L2)

In L1 the baseline top 3 were all from Section 9, including Section 9(4), which says that the parental consent rule does not apply in some prescribed cases. So Answer 1 was correct ("No, parental consent is not always required"). The cross-encoder ranked two general consent chunks from Section 6 higher, and the two exception chunks dropped to rank 5 and 7. Answer 2 then said that parental consent is required, without the exception. My guess is that the MS MARCO model mostly checks if the passage is about the same topic as the question and does not give much weight to the word "always".

In L2 the answer is in Section 13(3), which says a person must first "exhaust the opportunity of redressing her grievance" with the Data Fiduciary before going to the Board. The question uses "company" and "complain" while the Act uses "Data Fiduciary" and "grievance", and the chunk was not in the top 10 at all. Both answers were refusals, because the re-ranker can only reorder the 10 chunks it receives. (Because of the free API quota the two L2 answers came from different Gemini flash versions. Both refused, so it does not matter.)

5 Latency Profiling

I measured the retrieval part on the CPU (one thread) after a warm-up run. Each of the 10 queries was run 30 times, so there are 300 measurements for each stage. The LLM times are from the real API calls.

Table 4: Runtime per query in milliseconds

	Pipeline 1 (baseline)	Pipeline 2 (re-ranking)
Retrieval, mean	6.4	156.5
Retrieval, median	5.6	144.6
Retrieval, 95th percentile	10.6	234.7
LLM call, median of 10 queries	7459	6660
End-to-end, median of 10 queries	7466	6812

For retrieval the re-ranking pipeline is about 24 times slower than the baseline (6.4 ms vs. 156.5 ms, of which 150 ms is the cross-encoder). But in absolute terms it is still only 0.15 s per query. The LLM call takes about 7 s, so re-ranking adds only around 2% to the total time. Pipeline 2 even has a lower median end-to-end time, but that is only because the API response time changes a lot between requests. Some requests took much longer when Gemini was overloaded (the slowest one 240 s), so I used the median for the LLM rows.

6 Scaling Question

To see why we do not run the cross-encoder on the whole collection, I also ran it directly on the first n chunks of the document without the first stage (right side of Figure 2).

Table 5: Cross-encoder time when scoring n chunks per query

Chunks scored	Time per query	Time per pair
10	159 ms	15.9 ms
30	505 ms	16.8 ms
60	1011 ms	16.9 ms
120 (whole document)	2009 ms	16.7 ms

The time grows linearly with about 16.7 ms per (query, chunk) pair. For the whole document one query takes 2 s, which is 12.8 times slower than the two-stage pipeline, and this is a very small document. With the same rate, 10,000 chunks would need about 2.8 minutes per query, 1 million chunks about 4.6 hours, and 100 million chunks around 19 days.

The main reason is that the cross-encoder score depends on both the query and the chunk, so it cannot be computed before the query is known. Every new query needs one forward pass for every chunk in the collection. With a bi-encoder the chunk embeddings are computed only once, offline. At query time only the query is encoded and the index is searched, which took about 6 ms here. For large collections, approximate nearest neighbour indexes like HNSW or IVF keep this search fast even with millions of vectors. A GPU would make the cross-encoder faster per pair, but the time would still grow with the number of chunks.

So it makes sense to use both models: the bi-encoder to quickly get a short list of candidates, and the cross-encoder to sort only this short list more accurately. The weak point is that if the first stage misses the right chunk, the re-ranker cannot help (L2, and the Schedule chunk in F6).

7 Conclusion

On this document re-ranking fixed all 8 failure queries. The gold chunk moved from rank 4–8 into the top 3 and the LLM gave the correct answer instead of refusing, for about 0.15 s of extra time per query. There were also two problems. In L1 the cross-encoder pushed down an exception that the baseline had already found, and in L2 the right chunk was never retrieved by the first stage. Next I would try retrieving more candidates (for example 30 to 50, which would still take less than a second to re-rank), adding BM25 keyword search to the first stage, and testing a re-ranker trained on legal text.